

Faults in Capability Space

Study I — what 12 engine faults do to the attainable force and moment sets, and why that does not tell you which ones are survivable

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1 What this study does

Studies G, H and H-R ask whether a damaged vehicle can still fly a trajectory. This one asks the question underneath: **what can the damaged vehicle push and twist with at all**, before any trajectory is planned.

For an actuator set there are two sets in $\mathbb{R}^3$ and one in $\mathbb{R}^6$:

- the **attainable force set (AFS)** — every net force the actuators can produce;
- the **attainable moment set (AMS)** — every net moment;
- the **attainable wrench set (AWS)** — the 6-D set the other two are projections of, and the one an operating point has to lie inside.

Each RCS thruster is a one-sided segment: throttle $f \in [0, F]$ contributing $f d$ to the force and $f (r \times d)$ to the moment. Each gimbaled engine sweeps a spherical cap: thrust $T \in [T_{min}, T_{max}]$ along any direction within δ_{max} of its axis. The sum is what the vehicle can do instantaneously; its convex hull is what a support-function sweep recovers exactly, and that is what is computed here on 1200 directions per set.

The vehicle is the Study H plant: 2 gimbaled engines at $y_{eng} = \pm 0.25$ m with a 6° cone, 16 RCS thrusters at 445 N each, 7711 kg.

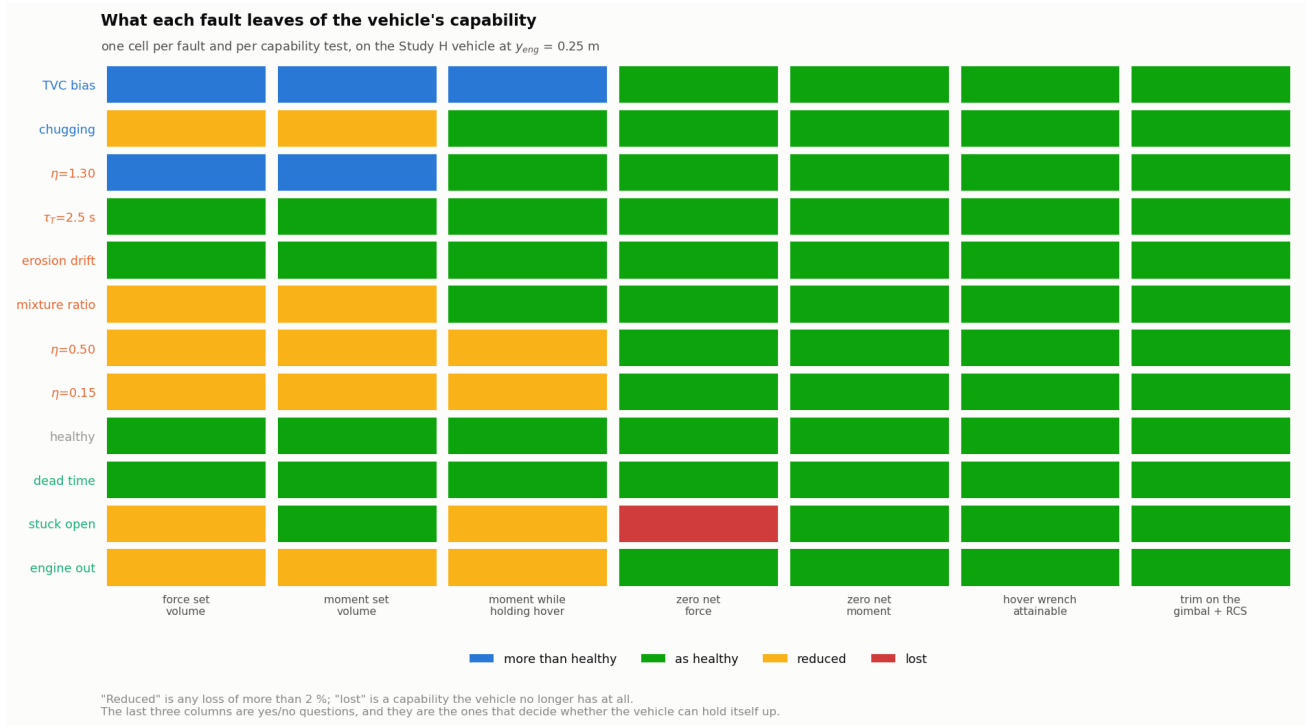


Figure 1: What each fault leaves of the vehicle's capability

2 Four things a fault can do to a set

The classification is decided from the numbers, not asserted:

Effect	Faults	What it means
shrink	chugging, $\eta=0.50$, $\eta=0.15$, mixture ratio, engine out	the damaged set is a subset of the healthy one
enlarge	TVC bias, $\eta=1.30$	more authority than healthy — and asymmetrically, which buys force and costs trim symmetry
deform	—	same size, moved: the set points somewhere else
puncture	stuck open	the origin leaves the set — the vehicle can no longer produce zero net wrench
nothing	healthy, $\tau_T=2.5$ s, erosion drift, dead time	the static set is identical to healthy; the fault lives entirely in time

2.1 Why the stuck-open valve is its own category

Everywhere else in this repository an engine can be shut down, so its minimum throttle does not shape the attainable set: zero is always available and the hull contains the origin. A valve stuck open removes exactly that. The engine's contribution becomes the shell $\{Td : T \in [T_{min}, T_{max}]\}$ with $T_{min} > 0$, whose hull does **not** contain the origin.

This is a qualitative change that no volume metric can see — the force set keeps 97% of its volume — and it is visible in the figure below as a set with its top cut off, the origin marked outside it.

Attainable force (top) and moment (bottom) sets under fault

Grey wireframe = the healthy set the damaged one is drawn inside. The dot is the origin — a red x marks a set that no longer contains it and so cannot produce zero net wrench.

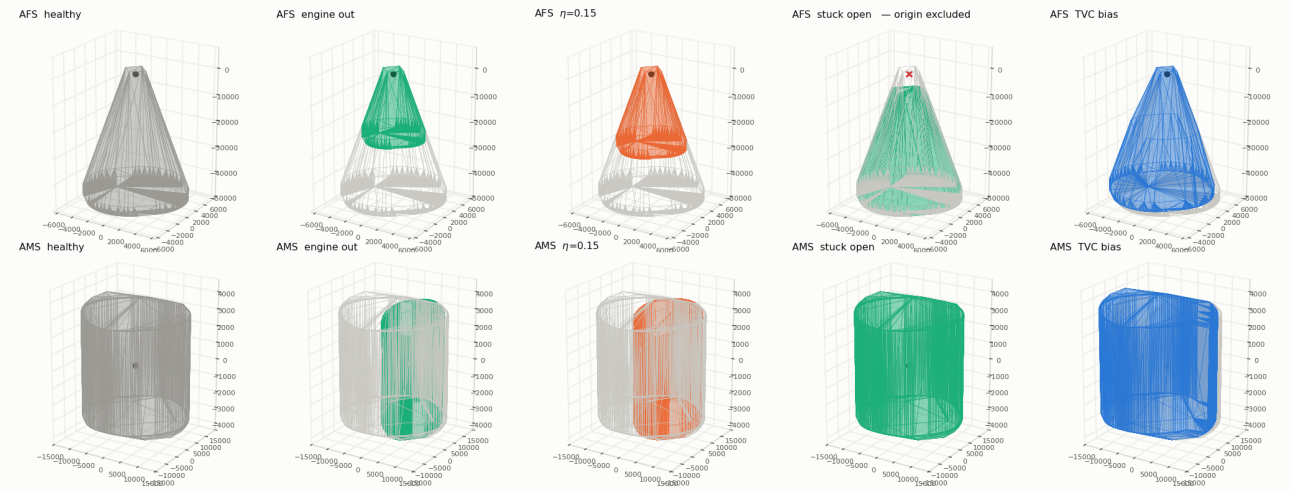


Figure 2: The attainable force and moment sets under fault

3 What each fault costs

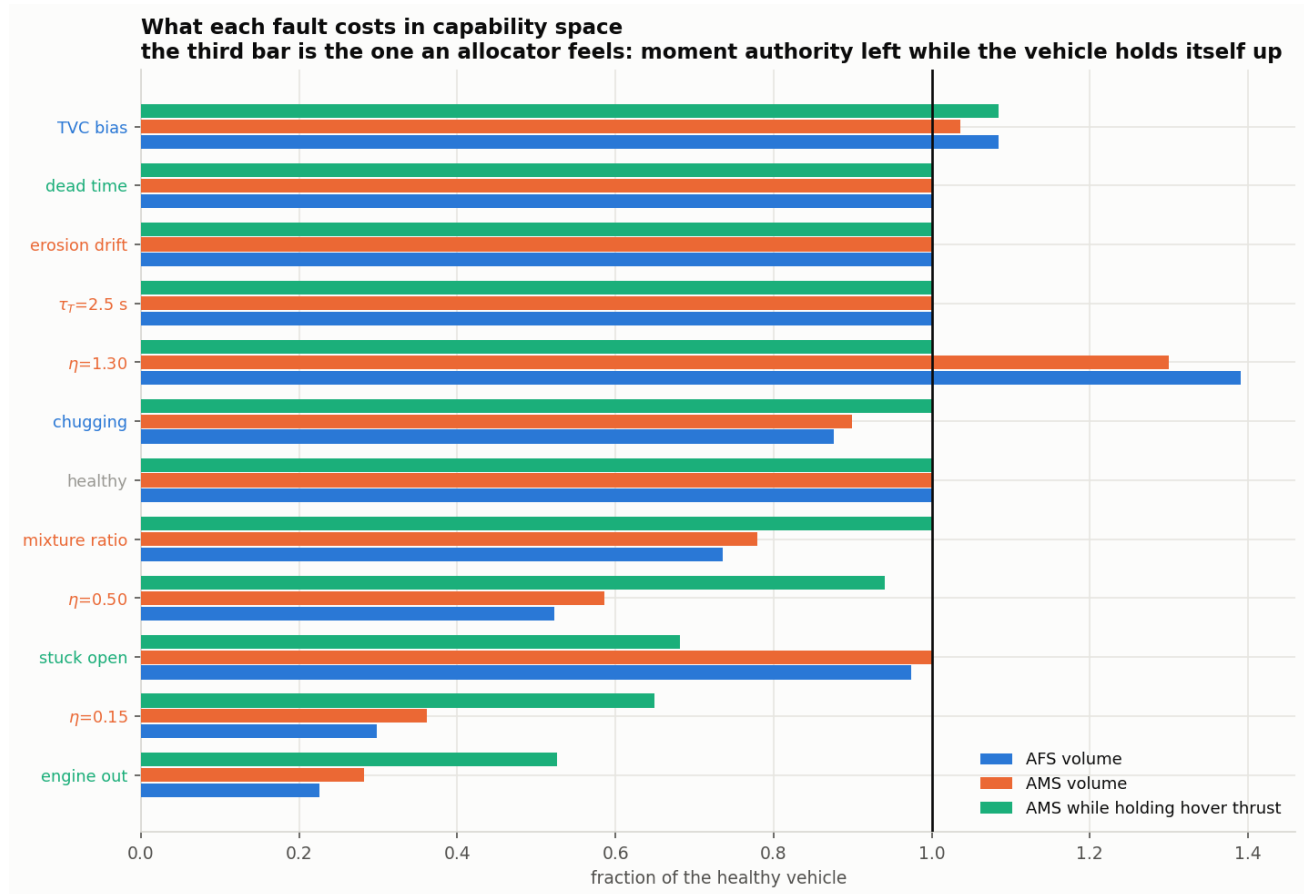


Figure 3: Capability cost per fault, in three measures

The third bar is the one that matters, and it is worth explaining why the first two are not enough.

The unconstrained AMS flatters every fault. It is free to spend the whole actuator set on one moment and let the vehicle fall. A descending vehicle cannot: most of its thrust is committed to not hitting the ground, and the moment authority that counts is what remains *after* that commitment. The conditional set — moments attainable while producing the 12530 N of vertical force that holds the vehicle up — is the set an allocator actually draws from, and it separates faults the raw AMS reports as nearly identical.

Fault	Effect	AFS	AMS	AMS	hover	Zero wrench	Hover wrench
engine out	shrunk	0.23×	0.28×	0.53×	yes	yes	73%
$\eta=0.15$	shrunk	0.30×	0.36×	0.65×	yes	yes	87%
stuck open	punctured	0.97×	1.00×	0.68×	no	yes	100%
$\eta=0.50$	shrunk	0.52×	0.59×	0.94×	yes	yes	93%
mixture ratio	shrunk	0.74×	0.78×	1.00×	yes	yes	100%
healthy	unchanged	1.00×	1.00×	1.00×	yes	yes	100%
chugging	shrunk	0.88×	0.90×	1.00×	yes	yes	100%
$\eta=1.30$	enlarged	1.39×	1.30×	1.00×	yes	yes	100%
$\tau_T=2.5$ s	unchanged	1.00×	1.00×	1.00×	yes	yes	100%
erosion drift	unchanged	1.00×	1.00×	1.00×	yes	yes	100%
dead time	unchanged	1.00×	1.00×	1.00×	yes	yes	67%
TVC bias	enlarged	1.08×	1.04×	1.08×	yes	yes	100%

4 Capability does not predict survivability

Every fault in the catalogue leaves the hover wrench attainable. Not one of them costs the vehicle the ability to hold itself up at this engine spacing. And yet Study H found four of them with injection points from which no trajectory exists at all.

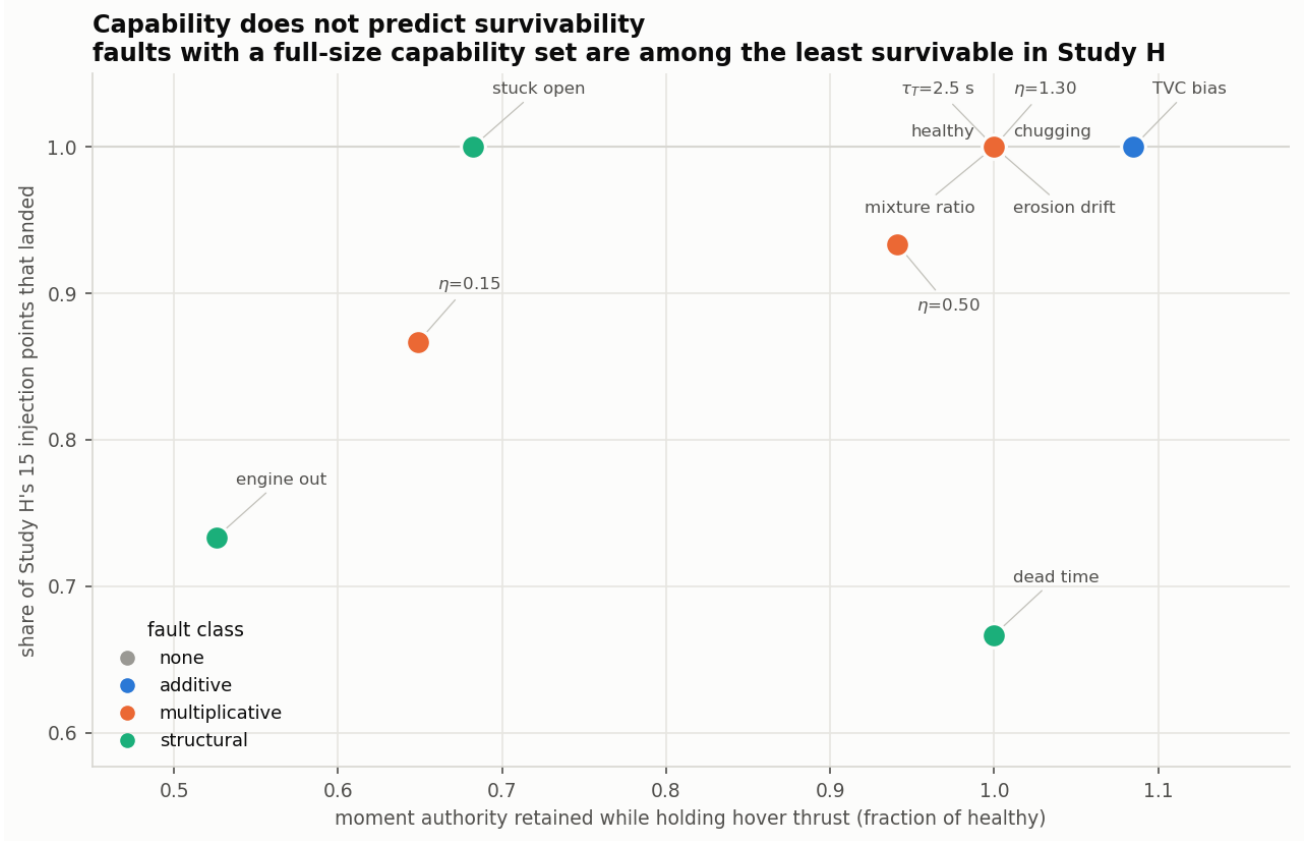


Figure 4: Retained moment authority against Study H landing share

The clearest case is **dead time**. Its capability sets are *identical* to healthy — same volume, same axes, same origin containment, same hover wrench — and it is the least survivable fault in the entire Study H campaign, landing from 67% of injection points and accounting for five of the nine injections that no relaxation of the trajectory constraints could recover.

A transport delay does not remove authority. It removes the vehicle’s ability to *apply* that authority when the state calls for it, and capability space — which is a statement about an instant, not about a sequence — cannot represent that. The same is true of the slow thrust response.

The converse case is $\eta=0.15$, which keeps only 65% of its conditional moment authority and still lands from 87% of injection points. Losing half the set is survivable when what remains still contains everything the trajectory asks for.

This is the study’s main result, and it is a negative one. An FTC scheme that reasons only about attainable sets — control-allocation feasibility, degraded-mode envelopes, reconfigurable allocation limits — will rank the temporal faults as harmless, because in capability space they are. Capability space is necessary and not sufficient: it tells you what the vehicle can do at an instant, and says nothing about whether it can do it *in time*.

The two measures are complementary, not redundant:

- a fault that leaves the required wrench outside the set is unrecoverable for a reason no controller can fix — that is a capability failure;
- a fault that leaves it inside may still be unrecoverable for reasons of timing, delay, or the state the vehicle happens to be in — that is a trajectory failure, and only a campaign like Study H finds it.

5 What redundancy buys

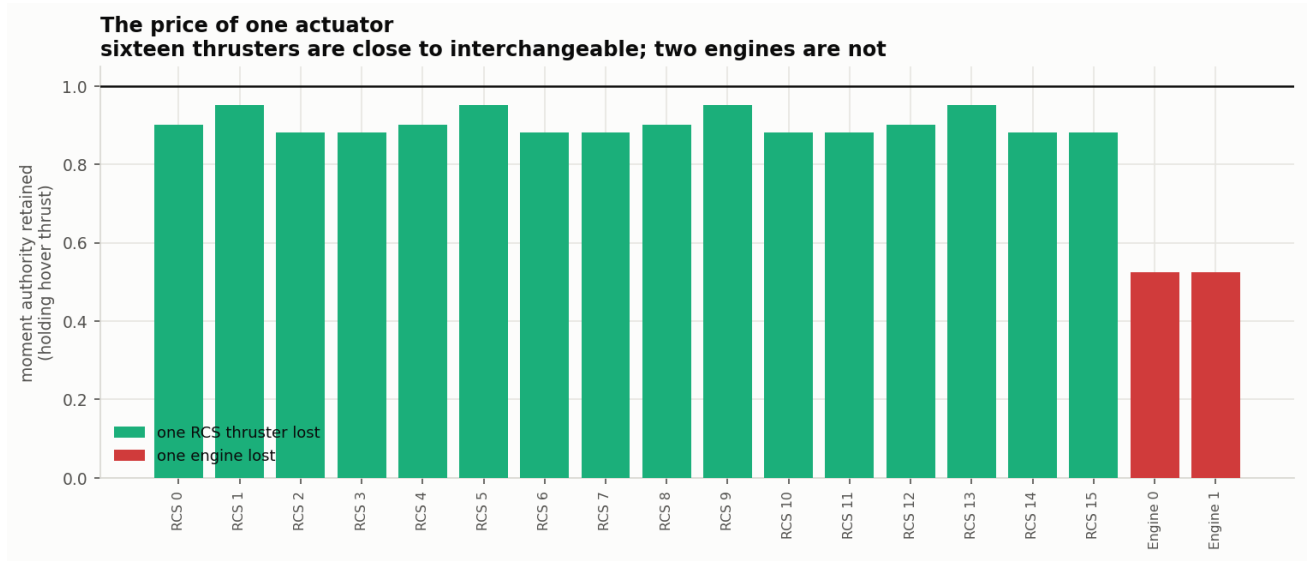


Figure 5: The cost of losing one actuator

Losing any single RCS thruster costs between 88% and 95% of the conditional moment authority — the worst of the sixteen (RCS 2) leaves 88%. Losing one of the two engines leaves 52%. **Sixteen thrusters are close to interchangeable; two engines are not**, and that asymmetry is the whole argument for treating the engine tier and the attitude tier as different redundancy problems rather than counting actuators.

5.1 The trim threshold, recovered from capability space

Study A derived the engine-out-trim limit analytically: a single gimbal can trim a one-engine-out asymmetry only while $y_{eng} \leq dz_{eng} \tan \delta_{max} = \mathbf{0.263\ m}$. That derivation is a statement about one moment balance. Asking capability space the same question — sweep the engine spacing, and for each ask the LP whether zero net moment is attainable while holding hover thrust — reproduces it without being told:

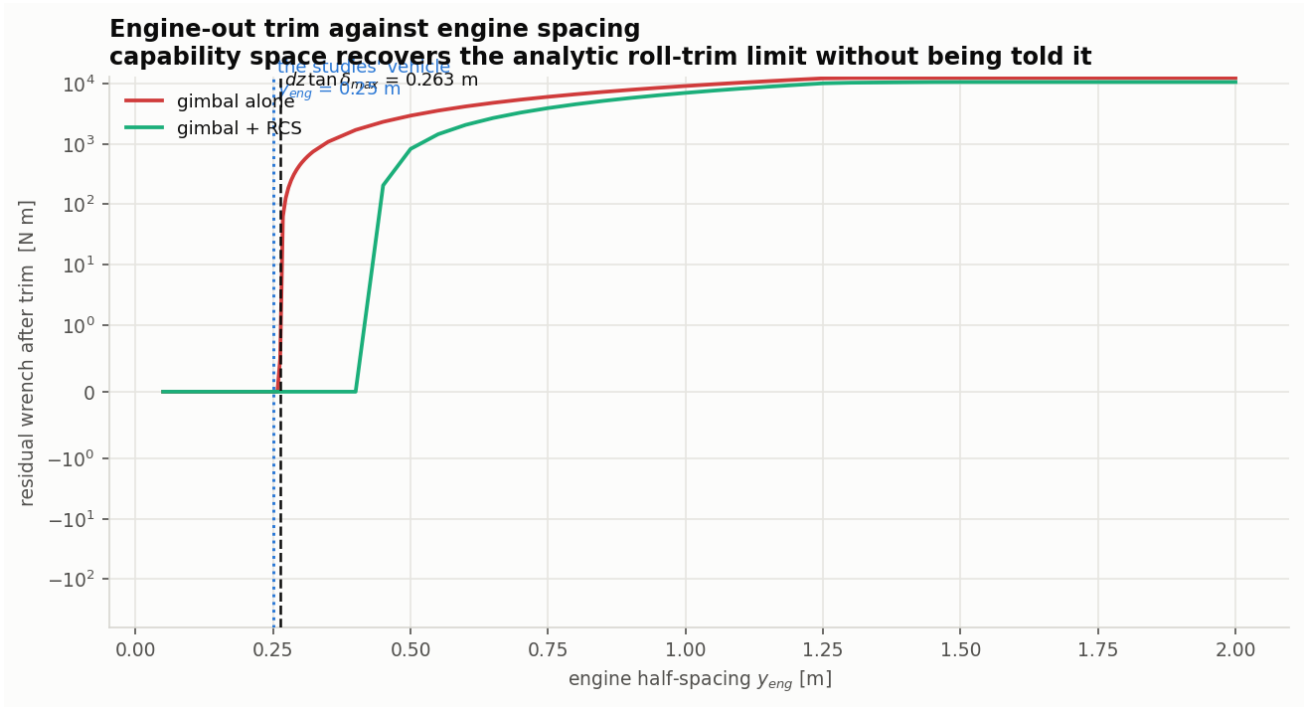


Figure 6: Engine-out trim against engine half-spacing

The gimbal-alone curve leaves zero residual up to $y_{eng} = 0.258 \text{ m}$ and diverges past it, against the analytic 0.263 m . The agreement is a check on the machinery: two independent derivations of the same threshold, one algebraic and one from a linear program over the actuator polytope.

The second curve is the part the analytic result does not cover. With the RCS tier available the vehicle trims engine-out at **every** spacing tested — the thrusters quietly cover what the gimbal cannot. That is worth knowing before concluding that a wide engine layout is disqualifying: it is disqualifying for the gimbal, not for the vehicle, as long as the attitude tier is healthy and has propellant. Study F's result that engine-out lands from 0 of 48 initial conditions at $y_{eng} = 1.5 \text{ m}$ is therefore not a capability statement either — the authority is there, and something else is spending it.

6 A fault that moves through capability space

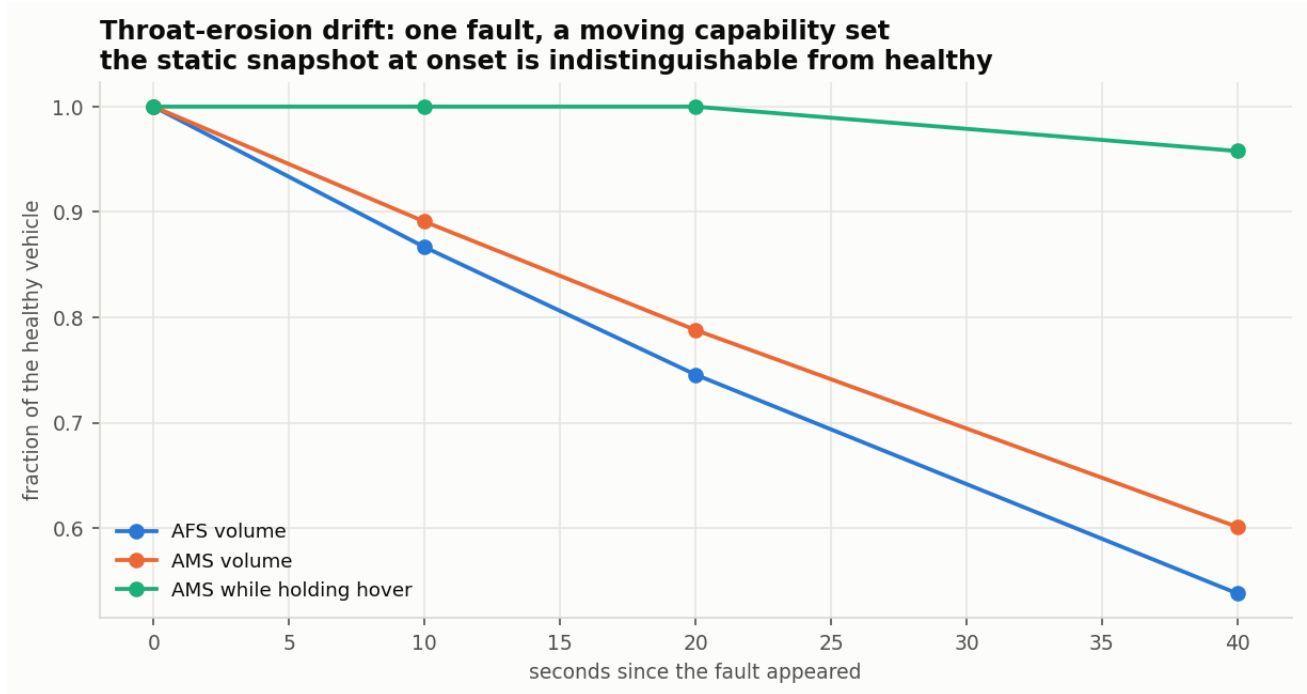


Figure 7: Throat-erosion drift over time

Throat-erosion drift is the one fault whose capability set is not a fixed object. Its efficiency decays from the moment of onset, so the set contracts continuously: the conditional moment authority falls from $1.00\times$ at onset to $0.96\times$ after 40 s.

At the instant of onset it is indistinguishable from healthy. Any capability-based monitor that samples the set will report a healthy vehicle for as long as it takes the drift to exceed the monitor’s threshold, and the fault that is easiest to detect in capability space is the one that has already done its damage. This is the argument for tracking the *rate* rather than the value.

7 Limitations

- **Instantaneous sets.** Everything here is a statement about one instant with no actuator dynamics. Rate limits, lags and delays are exactly what the unchanged-set faults consist of, which is the point of section 4 — but it means this study cannot be read as a survivability measure on its own.
- **Convex hulls.** The reported sets are hulls of the achievable set. For the RCS tier that is exact (a zonotope); for the engines the cap is sampled, and the LP uses an inscribed polytope, so a wrench reported attainable really is and a wrench reported unattainable is marginally conservative.
- **The chugging derate is a modelling choice.** A forced oscillation does not move the commanded boundary; what is reported is the authority the vehicle can hold regardless of ripple phase, which is the useful quantity but not the only defensible one.
- **One vehicle, one operating point.** The conditional sets are computed at hover thrust. A braking or pitch-over manoeuvre would commit the actuators differently and move every conditional number, though not the classification.
- **Fixed mass and inertia,** inherited from the plant.

8 Reproducing

```
python studies/capability_space/run_capability_study.py # the sets + LPs
python studies/capability_space/plot_capability.py      # figures
python studies/capability_space/build_report.py        # this document
```